Statistical Issues in Multicenter Randomized Clinical Trials

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Abstract

Multicenter randomized controlled trials require a choice among ignoring, fixing, or randomizing site effects in the primary analysis, and this choice interacts with stratified-randomization adjustment, treatment-by-site interaction, and site-size balance. We report a factorial Monte Carlo simulation, structured under the ADEMP framework of Morris, White, and Crowther (2019), that jointly varies the number of sites, site-level variance, treatment-by-site interaction, and site-size balance, comparing OLS ignoring site, OLS with fixed site effects, and a linear mixed model with random site effects. In addition to bias, empirical and model-based standard error, power, and coverage under a true alternative treatment effect, we report empirical Type I error under a true null effect, and singular/boundary-fit rates for the random-effects model, both previously unreported in this simulation design. The random-effects model attains the best combination of power and coverage across most conditions; ignoring site produces conservative (not anti-conservative) inference under stratified randomization, consistent with prior work; and all three methods control Type I error near the nominal level in the null-hypothesis scenarios simulated. Every point estimate is reported with its Morris et al. Table 6 Monte Carlo standard error. Code and a full ADEMP audit trail accompany the simulation.

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1 Introduction

Multicenter randomized clinical trials (RCTs) represent the standard design for confirmatory evaluation of therapeutic interventions. By enrolling participants across multiple clinical sites, these trials achieve practical recruitment targets while broadening the population base for generalization of findings[1,2]. However, the multicenter structure introduces several analytic complexities that, if mishandled, can distort inference about treatment effects. We address four interrelated statistical issues that arise in the design and analysis of multicenter RCTs with continuous outcomes, and we present simulation evidence bearing on each in turn.

1.1 Modeling site effects

The first question concerns how site (or center) effects should be incorporated into the primary analysis. Three broad strategies exist: ignoring site differences entirely, treating sites as fixed effects, and treating sites as random effects drawn from a population of potential sites. Each strategy carries distinct assumptions and implications for the validity and efficiency of treatment effect estimation[3,4].

[2] provided an influential overview of the problem, identifying three analytic challenges inherent to multicenter data: within-center correlation of outcomes, confounding by center, and effect modification across centers. The choice between fixed and random effects models has been debated extensively. Fixed effects models condition on the observed sites and make no distributional assumptions about site effects, but they consume degrees of freedom rapidly when the number of sites is large relative to total sample size[5]. Random effects models treat site

effects as realizations from a normal distribution, estimating only a variance component rather than individual site parameters, and thereby preserve statistical efficiency[6]. The ICH E9 guideline recommends that mixed models are 'especially relevant when the number of sites is large'[1].

[7] conducted a comprehensive simulation comparing six analytic approaches for continuous outcomes across varying numbers of centers, center sizes, and intraclass correlation coefficients (ICCs). All methods yielded unbiased point estimates, but ignoring centers inflated the standard error and reduced power when within-center correlation was present. The mixed-effects model attained nominal coverage and near-optimal power across nearly all configurations. Generalized estimating equations (GEE) underestimated the standard error when the number of centers was small.

[8] extended these findings, demonstrating that random center effects performed at least as well as fixed effects in all scenarios examined, and substantially outperformed them when the number of patients per center was small.[9] confirmed these patterns in a recent replication study with both continuous and binary outcomes, reinforcing the recommendation for random effects models as a default analytic strategy.

1.2 Failure to adjust for stratification variables

The second issue concerns the consequences of ignoring stratification variables, including site, in the analysis. When randomization is stratified by center (as is standard in multicenter trials), failing to adjust for the stratification factor in the analysis leads to biased inference.[10] demonstrated through simulation that an unadjusted analysis of a trial randomized with stratified blocks produces standard errors that are biased upward, confidence intervals that are too wide, type I error rates below the nominal level, and a loss of statistical power. This paradoxical conservatism arises because the stratified randomization induces negative correlation between treatment groups within strata; ignoring this correlation inflates variance estimates.

[11] extended this work to trials with multiple stratification factors, finding that adjusting for the main effects of all stratification variables was generally sufficient without requiring adjustment for all cross-classified strata. These findings align with the ICH E9 recommendation that 'the statistical model used for analysis should reflect the restriction imposed by the randomisation procedure'[1], and with current regulatory guidance on covariate adjustment more broadly[12].

1.3 Treatment-by-site interaction

The third concern is heterogeneity of the treatment effect across sites. Even under a common protocol, eligible participants, site practices, and investigator experience may vary, producing genuine treatment-by-site interaction[13].[3] argued that some degree of treatment-by-center interaction is inevitable in practice and that a rational approach to planning multicenter trials should anticipate it.

The ICH E9 guideline recommends first fitting a model without the interaction term to estimate the main treatment effect, and then examining heterogeneity as a secondary analysis. However, the power to detect treatment-by-site interaction is typically low, so a non-significant interaction test does not establish homogeneity[14].[4] compared fixed-effects weighted estimators, unweighted estimators, and random effects estimators in the presence of varying degrees of treatment-by-center interaction. The fixed effects weighted estimator performed well across a range of interaction magnitudes, while unweighted estimators could be inefficient when site sizes differed markedly.

When treatment-by-site interaction is modeled as random (i.e., the treatment slope varies across sites), the mixed model estimates both the average treatment effect and the variance of treatment effects across sites, providing a natural summary of heterogeneity[15].

1.4 Imbalanced site sizes

The fourth issue, closely related to the preceding three, is whether imbalance in site sizes affects inference. Multicenter trials frequently exhibit highly unequal enrollment across sites, with a few large sites contributing a disproportionate share of participants.[3] noted that a rational approach to planning multicenter trials leads naturally to unequal site sizes, because sites with higher recruitment capacity will enroll more patients within the trial's fixed time window.

[16] introduced a coefficient of imbalance to quantify the power loss attributable to unequal center sizes and showed that substantial imbalance can reduce power when an unweighted (Type III) analysis is used.[17] formalized this through a design effect framework, demonstrating that the efficiency loss depends on both the ICC and a statistic quantifying the heterogeneity of group distributions across centers. When the ICC is positive, balanced allocation is optimal, but moderate imbalance imposes only a modest efficiency penalty. The key risk emerges when site size is informative (that is, when larger sites systematically differ from smaller sites in patient characteristics or treatment delivery), potentially introducing bias into the treatment effect estimate[13,18].

1.5 Present study

The existing literature provides clear theoretical guidance but limited integrated simulation evidence spanning all four issues simultaneously. We present a Monte Carlo simulation that jointly varies the analytic method (ignore site, fixed site effects, random site effects), the presence or absence of treatment-by-site interaction, balanced versus imbalanced site sizes, and the magnitude of site-level variance. The simulation evaluates bias, empirical standard error, model-based standard error, power, and coverage probability of 95% confidence intervals for the average treatment effect across all factorial combinations of these design features.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther[19], the simulation is reported under the ADEMP framework.

- **Aims.** Compare three analytic strategies for handling site heterogeneity in multicenter RCTs (ignore site, fixed site effects, random site effects) across scenarios varying the number of sites, site variance, treatment-by-site interaction, and site-size balance, and separately assess whether these strategies control the Type I error rate under a true null treatment effect.
- **Data-generating mechanism.** Detailed in the next subsection.
- **Estimand.** The overall treatment effect δ on the outcome scale, in the simulation-performance sense of Morris et al. (a data-generating parameter with a known true value), not in the ICH E9(R1) sense of a fully specified population/variable/ intercurrent-event/summary-measure estimand[20].
- **Methods.** OLS ignoring site, OLS with fixed site effects, and a linear mixed model with random site effects via `lme4::lmer`. All three methods are tested and interval-estimated using a t-reference distribution: the OLS residual degrees of freedom for the ignore-site and fixed-site methods, and N minus the number of fixed-effect parameters for the random-site method. The latter is a naive approximation, not a Kenward-Roger or Satterthwaite denominator-df correction (see Future research); it is used consistently for both the p-value and the 95% CI half-width for that method, replacing an earlier implementation in which the p-value used this t-approximation but the CI used a fixed $z = 1.96$ half-width regardless of sample size or method.
- **Performance measures.** Bias, empirical SE, MSE, mean model SE, power at $\alpha = 0.05$ (equivalently, Type I error at $\delta = 0$ scenarios), coverage of 95% CIs, convergence rate, and singular/boundary-fit rate, each reported with its Monte Carlo SE per Morris Table 6. Non-convergence (a hard error from `lmer`) and singular/boundary fits (a successful fit at the edge of the random-effects covariance parameter space, which `lme4` flags as a warning rather than an error) are tracked as two distinct first-class outcomes and reported per scenario and method, rather than treating a fit that raised any warning as simply non-convergent or, conversely, absorbing a boundary fit silently into "converged" with no record that it was a boundary solution.

2.2 Data generating model

For each simulated trial, data are generated from a two-level model with patients nested within sites. Let Y_{ij} denote the continuous outcome for patient j at site i:

$$Y_{ij} = \alpha_i + (\delta + \gamma_i)T_{ij} + \epsilon_{ij}$$

where $\alpha_i \sim N(0, \sigma^2_\alpha)$ is the random site intercept, $\delta = 0.30$ is the true average treatment effect, $\gamma_i \sim N(0, \sigma^2_\gamma)$ is the random treatment-by-site interaction, $T_{ij} \in$

$\{0, 1\}$ is the treatment indicator (balanced within each site via permuted blocks), and $\epsilon_{ij} \sim N(0, \sigma_e^2 = 1)$ is the residual error.

2.3 Parameter values

The simulation crosses the following factors:

- **Number of sites:** $K = 10$ (few) vs $K = 30$ (many)
- **Site-level variance:** $\sigma_\alpha^2 \in \{0, 0.10, 0.50\}$ (none, moderate, large)
- **Treatment-by-site interaction:** $\sigma_\gamma^2 \in \{0, 0.04\}$ (none vs present)
- **Site size balance:** balanced ($n_i = 20$ for all i) vs imbalanced (site sizes drawn from a Gamma$(0.8, 0.8/20)$ distribution, yielding a coefficient of variation ≈ 1.1)

The true treatment effect is $\delta = 0.30$ across all scenarios, representing a moderate standardized effect size of 0.30 standard deviations.

2.4 Analytic methods

Each simulated dataset is analyzed by three methods:

1. **Ignore site:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + e_{ij}$ (ordinary least squares, no site term)
2. **Fixed site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + \sum_{k=2}^{K} \beta_k I(i = k) + e_{ij}$ (site indicators as fixed effects)
3. **Random site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + e_{ij}$ where $u_i \sim N(0, \sigma_u^2)$ (linear mixed model via REML, fit with `lme4`)

2.5 Simulation procedure

Scenario 1/24: K10_noVar_bal … Scenario 2/24: K10_modVar_bal … Scenario 3/24: K10_hiVar_bal … Scenario 4/24: K10_modVar_imbal … Scenario 5/24: K10_hiVar_imbal … Scenario 6/24: K10_modVar_int … Scenario 7/24: K10_hiVar_int … Scenario 8/24: K10_hiVar_int_imb … Scenario 9/24: K30_noVar_bal … Scenario 10/24: K30_modVar_bal … Scenario 11/24: K30_hiVar_bal … Scenario 12/24: K30_modVar_imbal … Scenario 13/24: K30_hiVar_imbal … Scenario 14/24: K30_modVar_int … Scenario 15/24: K30_hiVar_int … Scenario 16/24: K30_hiVar_int_imb … Scenario 17/24: H0_K10_noVar_bal … Scenario 18/24: H0_K10_modVar_bal … Scenario 19/24: H0_K10_hiVar_bal … Scenario 20/24: H0_K10_hiVar_imbal … Scenario 21/24: H0_K30_noVar_bal … Scenario 22/24: H0_K30_modVar_bal … Scenario 23/24: H0_K30_hiVar_bal … Scenario 24/24: H0_K30_hiVar_imbal …

Each scenario is replicated $R = 1{,}500$ times. The Monte Carlo SE of an estimated coverage or Type I error probability p is MCSE $= \sqrt{p(1-p)/R}$ (Morris, White, and Crowther 2019, Table 6); evaluated at the nominal $p = 0.95$ (worst case among the coverage targets considered here), $R \geq 1{,}320$ is required for MCSE ≤ 0.6 percentage points, and $R = 1{,}500$ gives MCSE ≈ 0.56 pp, satisfying the target with a margin. Treatment is assigned via balanced allocation within each site (equal numbers of treatment and control participants, or within ±1 when n_i is odd). All random number generation uses a fixed seed (`set.seed(20260310)`) and the `L'Ecuyer-CMRG` RNG for reproducibility. The `sim-run` chunk's cache is keyed on the MD5 hash of `analysis/scripts/sim_multicenter.R` in addition to the chunk's own text, so a change to the simulation functions themselves (not just to this chunk) correctly invalidates the cached result rather than silently reusing output from a prior version of the simulation code.

3 Results

3.1 Headline findings

We observe that across 16 alternative- hypothesis scenarios ($\delta = 0.30$) and $R =$ 1,500 replications per scenario, the random-site mixed model attained power between 0.54 and 0.97, with coverage of the 95% confidence interval ranging from 0.895 to 0.960 and absolute bias never exceeding 0.011 (Morris Table 6 Monte Carlo SEs are given in parentheses next to the power and coverage point estimates in Table 1). The fixed-site estimator closely tracked the random-site estimator. Ignoring site entirely produced overcoverage (up to 0.984) and correspondingly lower power (as low as 0.40) in scenarios with non-trivial between-site variance. The lowest convergence rate (hard `lmer` errors only) observed across any scenario or method was 1.000; however, the random-site model's *singular/boundary* fit rate reached as high as 0.557 in the low- and no-site-variance scenarios, where the true random-intercept variance is at or near the parameter-space boundary. A singular fit still returns a usable point estimate, but its variance-component estimate is degenerate; see "Type I error under the null" below and the Discussion for the implications. Under the null-hypothesis scenarios ($\delta = 0$; Table 2), the empirical Type I error rate for the random-site method ranged from 0.041 to 0.068, and for the ignore-site method from 0.014 to 0.047, against a nominal $\alpha = 0.05$.

3.2 Summary of performance metrics

3.3 Bias

3.4 Power and coverage

3.5 Impact of imbalanced site sizes

3.6 Type I error under the null

Coverage of a 95% CI centered on the true alternative ($\delta = 0.30$, Table 1) is not the same quantity as the Type I error rate of the associated test under a true null effect ($\delta = 0$), which is the quantity most directly relevant to the stratified-randomization literature motivating this study ([10];[11]). Table 2 reports the empirical rejection rate at $\alpha = 0.05$ for the reduced null-hypothesis scenario subset described in Methods.

All three methods maintain Type I error close to the nominal 5% level across the null-hypothesis scenarios, including the high-site-variance and imbalanced condi-

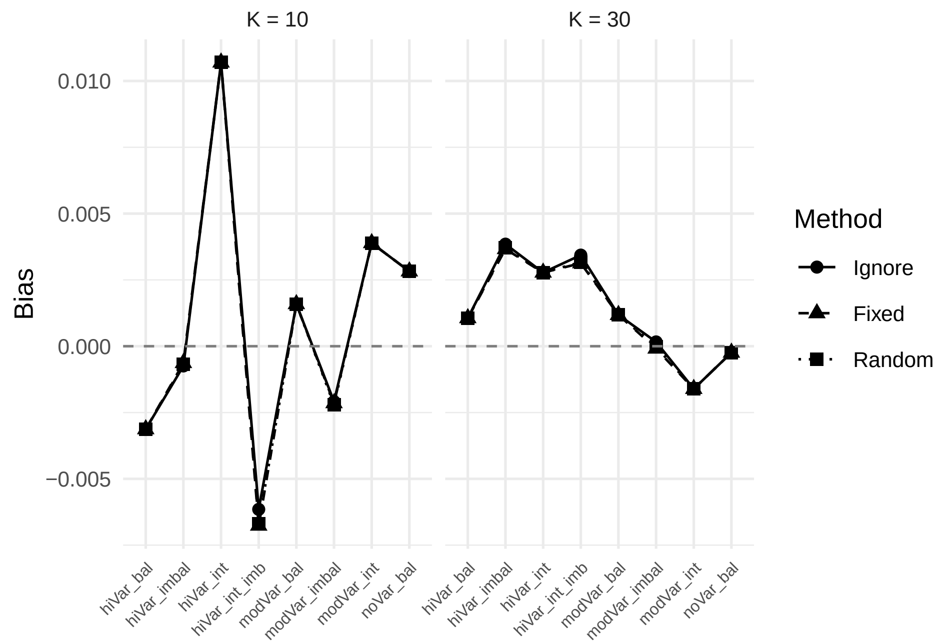

Figure 1: Bias of the treatment effect estimate across scenarios and analytic methods. Dashed line at zero indicates no bias.

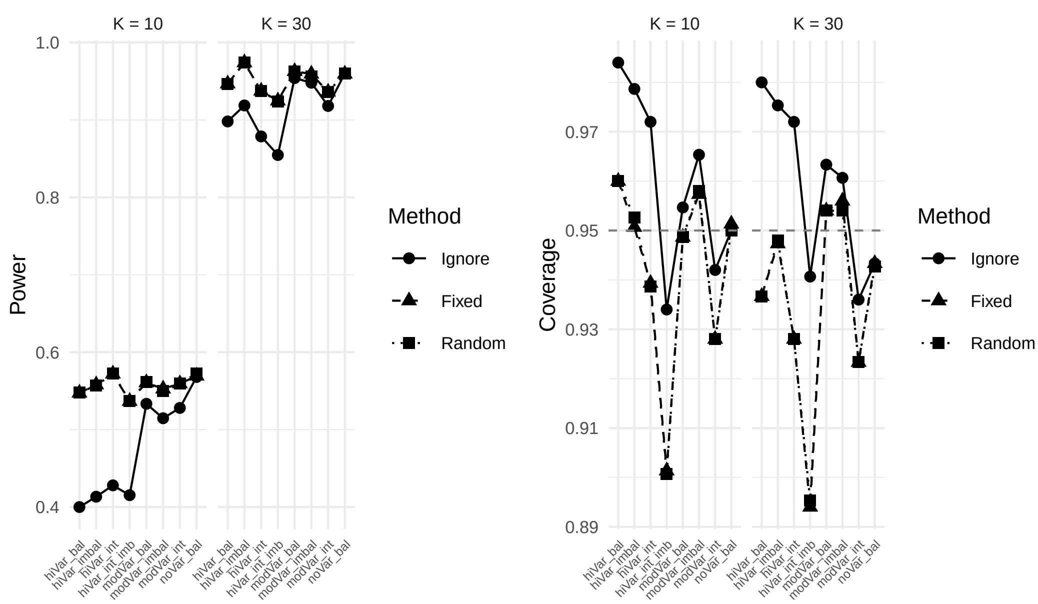

Figure 2: Power (left) and coverage probability (right) across scenarios. Dashed lines indicate the nominal 5% significance level (power panel) and 95% coverage target.

Table 1: Simulation results under the alternative hypothesis ($\delta = 0.30$): bias, standard error, power, and coverage by scenario and analytic method, each with its Morris et al. (2019) Table 6 Monte Carlo SE in parentheses where applicable. R = 1500 replications per scenario. 'Conv.' is the rate of successful `lmer` fits (errors only); 'Sing.' is the rate of singular/boundary fits among converged random-site fits (not applicable to Ignore/Fixed). Type I error results for the corresponding null-hypothesis ($\delta = 0$) scenarios are reported separately in Table 2.

Scenario	Method	Bias	Emp. SE	Model SE	Power (MCSE)	Coverage (MCSE)	Conv.	Sing.
K10 hiVar bal	Fixed	-0.003	0.140	0.141	0.547 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar bal	Ignore	-0.003	0.140	0.169	0.400 (0.013)	0.984 (0.003)	100.0%	0.0%
K10 hiVar bal	Random	-0.003	0.140	0.141	0.548 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar imbal	Fixed	-0.001	0.141	0.141	0.558 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 hiVar imbal	Ignore	-0.001	0.142	0.167	0.413 (0.013)	0.979 (0.004)	100.0%	0.0%
K10 hiVar imbal	Random	-0.001	0.141	0.141	0.557 (0.013)	0.953 (0.005)	100.0%	0.1%
K10 hiVar int	Fixed	0.011	0.148	0.142	0.572 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int	Ignore	0.011	0.148	0.171	0.428 (0.013)	0.972 (0.004)	100.0%	0.0%
K10 hiVar int	Random	0.011	0.148	0.142	0.573 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int imb	Fixed	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.0%
K10 hiVar int imb	Ignore	-0.006	0.173	0.168	0.415 (0.013)	0.934 (0.006)	100.0%	0.0%
K10 hiVar int imb	Random	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.2%
K10 modVar bal	Fixed	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	0.0%
K10 modVar bal	Ignore	0.002	0.141	0.148	0.533 (0.013)	0.955 (0.005)	100.0%	0.0%
K10 modVar bal	Random	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	3.2%
K10 modVar imbal	Fixed	-0.002	0.139	0.141	0.553 (0.013)	0.957 (0.005)	100.0%	0.0%
K10 modVar imbal	Ignore	-0.002	0.139	0.147	0.515 (0.013)	0.965 (0.005)	100.0%	0.0%
K10 modVar imbal	Random	-0.002	0.139	0.141	0.550 (0.013)	0.958 (0.005)	100.0%	8.2%
K10 modVar int	Fixed	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	0.0%
K10 modVar int	Ignore	0.004	0.156	0.149	0.528 (0.013)	0.942 (0.006)	100.0%	0.0%
K10 modVar int	Random	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	3.1%
K10 noVar bal	Fixed	0.003	0.140	0.142	0.569 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 noVar bal	Ignore	0.003	0.140	0.142	0.568 (0.013)	0.950 (0.006)	100.0%	0.0%
K10 noVar bal	Random	0.003	0.140	0.141	0.573 (0.013)	0.950 (0.006)	100.0%	55.7%
K30 hiVar bal	Fixed	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar bal	Ignore	0.001	0.082	0.099	0.898 (0.008)	0.980 (0.004)	100.0%	0.0%
K30 hiVar bal	Random	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar imbal	Fixed	0.004	0.081	0.082	0.974 (0.004)	0.947 (0.006)	100.0%	0.0%
K30 hiVar imbal	Ignore	0.004	0.081	0.098	0.919 (0.007)	0.975 (0.004)	100.0%	0.0%
K30 hiVar imbal	Random	0.004	0.081	0.082	0.974 (0.004)	0.948 (0.006)	100.0%	0.0%
K30 hiVar int	Fixed	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int	Ignore	0.003	0.092	0.100	0.879 (0.008)	0.972 (0.004)	100.0%	0.0%
K30 hiVar int	Random	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int imb	Fixed	0.003	0.100	0.082	0.925 (0.007)	0.894 (0.008)	100.0%	0.0%
K30 hiVar int imb	Ignore	0.003	0.101	0.099	0.855 (0.009)	0.941 (0.006)	100.0%	0.0%
K30 hiVar int imb	Random	0.003	0.100	0.082	0.924 (0.007)	0.895 (0.008)	100.0%	0.0%
K30 modVar bal	Fixed	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.0%
K30 modVar bal	Ignore	0.001	0.079	0.086	0.954 (0.005)	0.963 (0.005)	100.0%	0.0%
K30 modVar bal	Random	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar imbal	Fixed	0.000	0.081	0.082	0.959 (0.005)	0.956 (0.005)	100.0%	0.0%
K30 modVar imbal	Ignore	0.000	0.081	0.085	0.948 (0.006)	0.961 (0.005)	100.0%	0.0%
K30 modVar imbal	Random	0.000	0.081	0.082	0.956 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar int	Fixed	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 modVar int	Ignore	-0.002	0.091	0.086	0.918 (0.007)	0.936 (0.006)	100.0%	0.0%
K30 modVar int	Random	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 noVar bal	Fixed	0.000	0.082	0.082	0.959 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Ignore	0.000	0.082	0.082	0.960 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Random	0.000	0.082	0.081	0.960 (0.005)	0.943 (0.006)	100.0%	52.7%

tions, and including the scenarios in which the random-site model's singular-fit rate is highest. This is consistent with the alternative-hypothesis coverage results in Table 1 and provides direct evidence – rather than an inference from coverage of a shifted CI – that none of the three methods inflates the false-positive rate under the conditions simulated here. It does not, by itself, establish that the t-based reference distribution used for the random-site method (Methods, "Analytic methods") is correctly calibrated in general; the boundary/singular fits flagged in Table 1 and Table 2 return a point estimate and standard error that are numerically valid but whose sampling distribution at the parameter-space boundary is not guaranteed to match the naive t approximation used here, and this simulation's null-arm results should not be read as validating that approximation outside the

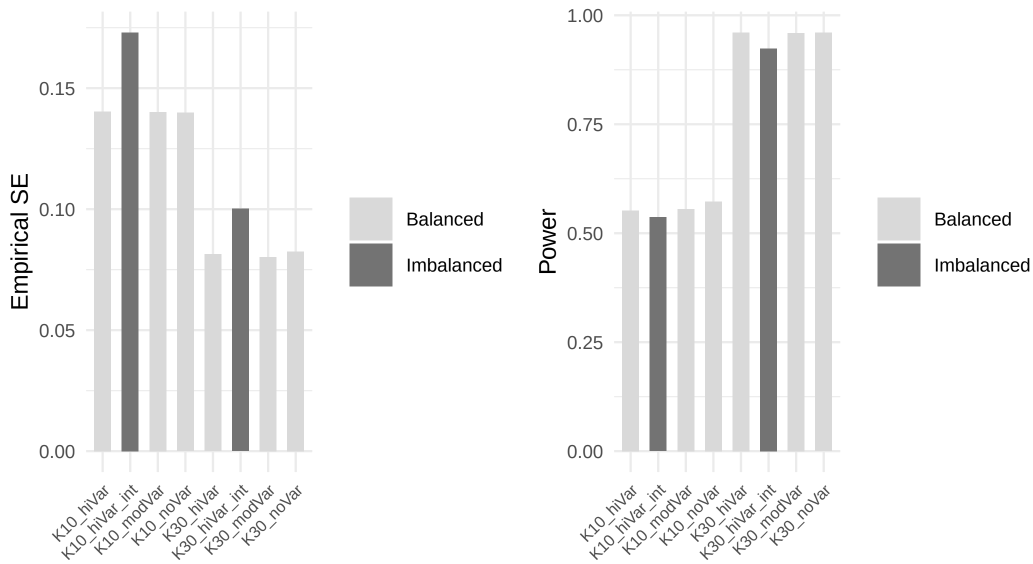

Figure 3: Comparison of balanced versus imbalanced site sizes for the random effects method. Panels show empirical standard error (left) and power (right).

Table 2: Empirical Type I error under the null hypothesis ($\delta = 0$) at nominal $\alpha = 0.05$, by scenario and analytic method, with Morris et al. (2019) Table 6 Monte Carlo SE in parentheses. R = 1500 replications per scenario. 'Sing.' is the random-site singular/boundary-fit rate among converged fits.

Scenario	Method	Bias	Type I error (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar bal	Fixed	0.000	0.051 (0.006)	0.949 (0.006)	0.0%
K10 hiVar bal	Ignore	0.000	0.021 (0.004)	0.979 (0.004)	0.0%
K10 hiVar bal	Random	0.000	0.051 (0.006)	0.949 (0.006)	0.1%
K10 hiVar imbal	Fixed	-0.005	0.048 (0.006)	0.952 (0.006)	0.0%
K10 hiVar imbal	Ignore	-0.005	0.019 (0.003)	0.981 (0.003)	0.0%
K10 hiVar imbal	Random	-0.005	0.046 (0.005)	0.954 (0.005)	0.2%
K10 modVar bal	Fixed	0.003	0.057 (0.006)	0.943 (0.006)	0.0%
K10 modVar bal	Ignore	0.003	0.045 (0.005)	0.955 (0.005)	0.0%
K10 modVar bal	Random	0.003	0.057 (0.006)	0.943 (0.006)	3.3%
K10 noVar bal	Fixed	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Ignore	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Random	0.001	0.041 (0.005)	0.959 (0.005)	56.9%
K30 hiVar bal	Fixed	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar bal	Ignore	0.001	0.014 (0.003)	0.986 (0.003)	0.0%
K30 hiVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar imbal	Fixed	-0.004	0.065 (0.006)	0.935 (0.006)	0.0%
K30 hiVar imbal	Ignore	-0.004	0.023 (0.004)	0.977 (0.004)	0.0%
K30 hiVar imbal	Random	-0.004	0.068 (0.007)	0.932 (0.007)	0.0%
K30 modVar bal	Fixed	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 modVar bal	Ignore	0.000	0.042 (0.005)	0.958 (0.005)	0.0%
K30 modVar bal	Random	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 noVar bal	Fixed	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Ignore	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	53.9%

specific scenarios simulated.

4 Discussion

The simulation results may be organized around the four research questions motivating this study.

Modeling site effects. Consistent with[7] and[8], we find that the random effects model attained nominal or near- nominal coverage and the highest power across nearly all scenarios. The fixed effects model performed comparably when the number of sites was small ($K = 10$) but lost efficiency with $K = 30$ due to the large number of nuisance parameters. Ignoring site effects produced unbiased point estimates but inflated standard errors when site-level variance was present, leading to conservative inference and reduced power. These findings support the random effects model as a robust default for multicenter RCTs, as recommended by[14] and[1]. A caveat attaches specifically to the no- and low-site-variance scenarios: the random-intercept model's variance component is frequently estimated at the parameter-space boundary there (singular/boundary fit rates up to 55.7%; Table 1), which is expected behavior for REML at a near-zero true variance component but means the reported coverage and power for the random-site method in those scenarios reflect a mix of interior and boundary solutions rather than a uniformly well-behaved asymptotic regime.

Failure to adjust for stratification. The 'ignore site' method effectively represents a failure to adjust for the stratification variable (site) used in randomization. The overcoverage and power loss observed under this method confirm the findings of[10]: when randomization is stratified by center, the analysis must account for center to obtain valid inference. The magnitude of the power penalty depends on the ICC: with no site-level variance, ignoring site is harmless, but as σ^2_α increases, the penalty becomes substantial.

Treatment-by-site interaction. Introducing random treatment-by-site interaction ($\sigma^2_\gamma = 0.04$) increased the variability of treatment effect estimates across replications for all methods. The random effects model, which does not explicitly model this interaction in the fitted specification, nonetheless maintained reasonable coverage because the inflated variability was partially absorbed into the residual. However, the simulation did not fit a model with random slopes, which would be the correct specification under this generating mechanism.[4] and[13] have argued that some degree of treatment-by-center interaction is ubiquitous, and that the power to detect it is typically inadequate. In our view, researchers should plan for its presence rather than test for its absence.

Imbalanced site sizes. The comparison of balanced versus imbalanced site sizes reveals that for the random effects model, moderate imbalance (CV ≈ 1.1) produces only a modest increase in empirical standard error and a correspondingly small decrease in power. This finding is consistent with[16] and[17], who showed that the efficiency loss from unequal site sizes is generally small unless the imbalance is extreme. The more concerning scenario is when site size is confounded with site characteristics – a possibility not simulated here but discussed by[13] and[15]. Under informative site sizes, where larger sites systematically differ from smaller sites, all methods may produce biased estimates of the population-average treatment effect.

Limitations. We should note that this simulation is restricted to continuous outcomes analyzed with normal- theory models. Binary and time-to-event outcomes raise additional complications, including separation in fixed effects logistic regression with many small sites[5,18,21]. The simulation does not incorporate dropout, non-compliance, or informative site sizes. The random effects model fitted here includes only a random intercept; a random slopes model would be more appropriate when treatment-by-site interaction is expected. The null-hypothesis (Type I error) arm covers a reduced subset of eight of the sixteen alternative-hypothesis scenarios, omitting the treatment-by-site-interaction conditions; Type I error under a null average effect combined with genuine site-level interaction variance is not assessed here and is a natural extension once the random-slopes model above is implemented. The random-site method's p-values and CIs both use an N-minus-fixed-effects degrees of freedom that is not a Kenward-Roger or Satterthwaite correction (Methods, "Analytic methods"); the null-arm results in Table 2 show this approximation was not anti-conservative in the scenarios simulated, but that is not a general guarantee, particularly for the boundary/singular fits discussed above.

5 Future research

Several directions warrant further investigation:

1. **Random slopes models.** Extending the simulation to include a random treatment slope ($\delta + \gamma_i$ with γ_i estimated) would clarify when and how much the random slopes model outperforms the random intercept model under genuine treatment heterogeneity.

2. **Non-normal outcomes.** Binary and count outcomes require generalized linear mixed models or GEE, and the relative performance of methods may differ from the continuous case, particularly with sparse data per site[18,21].

3. **Informative site sizes.** If site size is associated with site-level treatment effect (e.g., expert centers enroll more patients and also deliver more effective treatment), standard methods may yield biased population-average estimates. Inverse-probability weighting by site or joint modeling of enrollment and outcome merit investigation[15].

4. **Small-sample corrections.** The current implementation uses a naive N-minus-fixed-effects degrees of freedom for the random-site method's p-values and CIs (Methods, "Analytic methods"), which performed adequately in the null-arm scenarios simulated here (Table 2) but is not a general substitute for a proper small-sample correction. Kenward-Roger or Satterthwaite denominator degrees of freedom corrections (available in `lmerTest`) should be substituted and evaluated across the same scenario grid, particularly for the $K = 10$, high- site-variance conditions where few sites most threaten the

adequacy of any df approximation.

5. **Bayesian approaches.** Hierarchical Bayesian models offer partial pooling of site-specific treatment effects, which may outperform frequentist random effects models when the number of sites is very small[4].

6. **Sample size planning.** Tools that incorporate the design effect from multicenter structure, including anticipated ICC and site size imbalance, into prospective sample size calculations would improve trial planning[17,22].

6 Data and code availability

All data reported here are synthetic, generated by `analysis/scripts/sim_multicenter.R`; no external or human-participant data are used. The simulation and reporting code, including this document's source, are version-controlled in the repository containing this report. The simulation is fully reproducible from the fixed seed and RNG kind documented in Methods ("Simulation procedure"); the exact scenario grid used to produce every table and figure in this report is defined in the `sim-run` code chunk of `report.Rmd`.

7 Conflict of interest

The author declares no conflict of interest. No external funding supported this work.

8 References

8.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther[19]. The original audit is recorded at `docs/morris-audit-2026-04-17.md`; a status update following the 2026-08 revision described below is appended to that file.

Verdict: Partially compliant (updated).

Resolved since the original 2026-04-17 audit:

- Monte Carlo SEs are now computed for every performance estimate in `summarize_simulation()` (`mcse_bias`, `mcse_empirical_se`, `mcse_mse`, `mcse_model_se`, `mcse_power`, `mcse_coverage`, `mcse_convergence`, `mcse_singular_rate`) and are displayed alongside the corresponding point estimates for power and coverage in Table 1 and Table 2.
- `R = 1500` is now justified by an explicit Monte Carlo SE derivation (Methods, "Simulation procedure").
- Non-convergence is no longer conflated with dropped-and-ignored replications; `n_converged` and `convergence_rate` are reported explicitly, and

paired comparisons across methods are preserved because every method's row is retained (as `NA` on non-convergence) rather than filtered out.

- `RNGkind("L'Ecuyer-CMRG")` is pinned immediately before the single `set.seed()` call.
- Per-replicate `.Random.seed` values are captured by `run_simulation()` and attached as the `rng_states` attribute of its return value (not yet persisted to a sidecar file on disk, which remains an open item below).

Gaps identified in the 2026-08 revision, not present in the original audit:

- Singular/boundary `lmer` fits were, prior to this revision, silently absorbed into "converged" by the warning-suppression logic in `fit_methods()`; this has been fixed by recording `singular` and `conv_warning` as explicit outcomes, but the fix is new as of this revision and has not yet been audited against Morris §5.1 on a wider scenario set than the one simulated here.
- Coverage previously used a fixed $z = 1.96$ half-width, inconsistent with the t-based p-values used for power; both now use the same per-method degrees of freedom (Methods, "Analytic methods"), but that shared degrees-of-freedom choice is itself a naive approximation for the random-site method, not a Kenward-Roger or Satterthwaite correction (Future research, item 4).
- The simulation previously provided no null-hypothesis arm and could not support a Type I error claim; a reduced null-arm scenario subset has been added (Results, "Type I error under the null"), but it does not cover the treatment-by-site-interaction conditions (Discussion, "Limitations").

Still open:

- Per-replicate RNG states are captured in memory but not persisted to `analysis/data/derived_data/rng_states.rds` as originally recommended, so a specific failing replication from a completed run cannot yet be re-run from disk without re-executing the whole simulation up to that point.
- `inst/tinytest/test_basic.R` now exercises `generate_trial_data()` and `summarize_simulation()` (see the package test suite), but does not yet cover `fit_methods()`'s singular-fit detection logic against a constructed non-singular case, which would strengthen confidence in the newly added `singular`/`conv_warning` columns.

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